Drawdowns – Minimizing Variables is Crucial in Achieving Reliable Test Results

When performing any physical testing of liquid coatings, minimizing variables is an important prerequisite for achieving accurate, reliable and repeatable test results, whether you are evaluating color, gloss, film thickness, adhesion or some other property. This is especially true when making drawdowns, a fist step in many test procedures.

Drawdowns typically require:

- A standardized substrate (chart or card)
- A level, rigid support surface for the standardized substrate
- A device (applicator) to ensure consistency of coating application at a controlled film thickness

Standardized Substrates and Important Chart Characteristics

Although a variety of materials, such as plastic, metal and wood can be used, a sealed paper drawdown chart or card, commonly with various black and white patterns, is usually the substrate of choice. To achieve the goal of minimizing variables, it is important that certain characteristics are considered when selecting a chart. These include:

- **Color uniformity:** Consistency of color and reflectance over the white and black areas are particularly important when working with semi-transparent or clear coatings because show-through of the substrate can affect color measurements of the applied coating.
- **Imperviousness/Solvent resistance:** The chart's surface needs to be sealed with a clear topcoat which acts as a strong barrier to prevent water molecules from penetrating its surface and solvents from attacking it, thereby ensuring that smooth, uniform drawdowns are produced. If you see such failures as tiny bumps or eruptions, "crazing" or cracking in the applied coating, it is clear evidence that the chart is inadequately sealed.

- **Wettability:** The chart's surface needs to be compatible with most water and solvent based coatings. If the clear topcoat is not formulated with this requirement in mind, then applied coatings may "crawl", show "fish eyes" or other characteristics of poor wetting.
- **Surface smoothness:** The smoother and the less bumpy the chart surface is, the better the appearance of your drawdown. If the chart has a bumpy surface, the uneven texture may be telegraphed through the drawdown, resulting in a distinct orange-peel appearance.
- **Curl resistance:** The chart should be flexible yet sufficiently sturdy so that it will lie flat both before and after application of your coating. Lower quality charts may curl due to moisture in the air or water vapor transmission occurring during applications of water-based paints.
- **Non-fluorescence (no optical brighteners):** The chart should be free of optical brighteners. The presence of optical brighteners can cause misleading results when making instrumental or visual color evaluations.

Suprt Surface

The surface which supports the drawdown chart is also critical. A desk or table top should not be presumed to be flat as it may have wear or distortion which may result in coating thickness variations. The optimal surface for supporting drawdown charts is a glass plate because of its superior planarity (see Leneta Item GDP-1 Glass Drawdown Plate). In addition to glass drawdown plates, vacuum plates are frequently used to ensure that the chart is held flat (see Leneta VP-0912 Vacuum Plate).

Aplicator

Straight bar or blade type applicators are the most frequently used devices for drawing down paints and related coatings (see Bird Applicators). Wire wound rods (see Wire-Cators) are also popular due to their much lower cost as well as their ability to apply edge-to-edge drawdowns while producing films with good consistency.